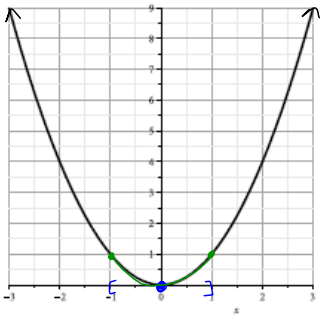
Section 5.5 - Absolute Extrema

Definition: An **absolute maximum** is the largest value a function obtains on its domain. An **absolute minimum** is the smallest value a function obtains on its domain.

Example 1: Find the absolute maximum and absolute minimum of each function if they exist.

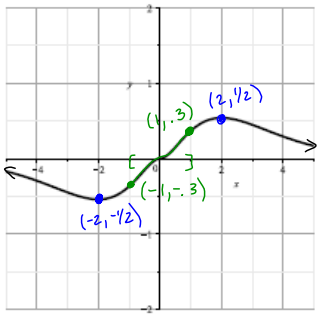
a)



No Absolute Maximum

Absolute Minimum value of 0 occurs at $x=0$

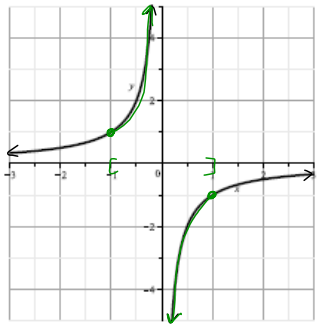
b)



Abs Max of 1/2 occurs at $x=2$

Abs Min of $-1/2$ occurs at $x=-2$

c)



No Abs Min

No Abs Max

Extreme Value Theorem: A function f that is continuous on a closed interval $[a,b]$ has both an absolute maximum value and an absolute minimum value on that interval.

Example 2: Find the absolute max and min for the functions in Example 1 on the interval $[-1,1]$

a) Abs Max of 1 occurs at $x=-1$ and $x=1$
Abs Min of 0 occurs at $x=0$

b) Abs Max of .3 occurs at $x=1$
Abs Min of $-.3$ occurs at $x=-1$

c) No Abs Max or Min

Finding Absolute Extrema on a Closed Interval

1. Check to make certain that f is continuous over $[a,b]$
2. Find the critical values in the interval (a,b).
3. Evaluate f at the endpoints a and b and at the critical values found in step 2.
4. The largest value obtained from the previous step is the absolute maximum of $f(x)$ on $[a,b]$ and the smallest value obtained is the absolute minimum of $f(x)$ on $[a,b]$.

Example 3: Find the absolute extrema of the following functions on the given intervals

a) $f(x) = 2x^3 - 3x^2 - 12x + 24$ on $[1,4]$ always continuous

$f'(x) = 6x^2 - 6x - 12$

$= 6(x^2 - x - 2)$

$= 6(x-2)(x+1)$

$f'(x) = 0$ when

$x - 2 = 0$ $\quad$ $x + 1 = 0$

$x = 2$ $\quad$ $\cancel{x = -1}$

only CV on our interval $\quad$ not on our interval

x	$f(x)$
1	11
2	4
4	56

- An abs max of 56 occurs at $x=4$
- An abs min of 4 occurs at $x=2$

always continuous

b) $f(x) = x^3 + 3x^2 - 9x - 7$ on $[-2,0]$

$f'(x) = 3x^2 + 6x - 9$

$= 3(x^2 + 2x - 3)$

$f'(x) = 0$ when

$3(x^2 + 2x - 3) = 0$

$3(x+3)(x-1) = 0$

$x + 3 = 0$ $\quad$ $x - 1 = 0$

$\cancel{x = -3}$ $\quad$ $\cancel{x = 1}$

neither are on our interval

x	$f(x)$
-2	15
0	-7

Abs Max of 15 at $x=-2$

Abs Min of -7 at $x=0$

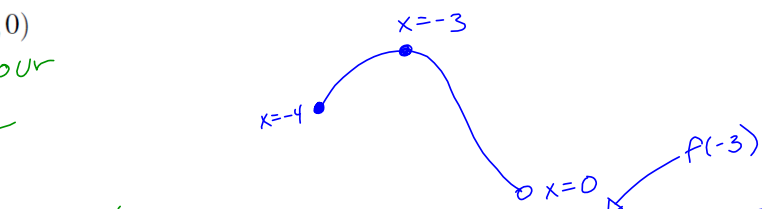
c) $f(x) = x^3 + 3x^2 - 9x - 7$ on $[-4,0)$

From part b, we found our critical values to be $x = -3$ and $\cancel{x = 1}$

the only critical value on that interval

$f''(x) = 6x + 6$

$f''(-3) = 6(-3) + 6 = -$



An absolute max of 20 at $x=-3$.

No absolute min

Second Derivative Test for Absolute Extrema For a continuous function, f, on any interval I (open, closed, or half-open), if $x = c$ is the only critical value in the interval where $f'(c) = 0$ and if $f''(c)$ exists, then $f(c)$ is

- an absolute minimum on I if $f''(c) > 0$
- an absolute maximum on I if $f''(c) < 0$

Note: The test fails if $f''(c) = 0$.

Example 4: Find the absolute extrema of each function on the given interval.

always continous

a) $f(x) = 6x - x^2 + 4$ on $(0, \infty)$

$f'(x) = 6 - 2x$

$f'(x) = 0$ when

$6 - 2x = 0$

$6 = 2x$

$3 = x$

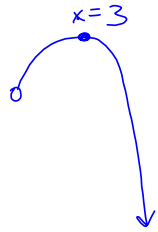
only CV on our interval

$f''(x) = -2$

$f''(3) = -2$

An abs max of 13 at $x = 3$.

NO abs min



cont. on our interval

b) $f(x) = 2x + \frac{8}{x}$ on $(0, 10)$

$= 2x + 8x^{-1}$

$f'(x) = 2 - 8x^{-2}$

$= 2 - \frac{8}{x^2}$

$= \frac{2x^2}{x^2} - \frac{8}{x^2} = \frac{2x^2 - 8}{x^2}$

$f'(x) = 0$ when

$2x^2 - 8 = 0$

$2x^2 = 8$

$\sqrt{x^2} = \sqrt{4}$

$x = 2, \cancel{x = -2}$

only CV on our interval

not on our interval

$f'(x)$ is undefined when $x^2 = 0$

$\cancel{x = 0}$

Not a CV since not in domain of $f(x)$

$f''(x) = 16x^{-3}$

$= \frac{16}{x^3}$

$f''(2) = \frac{16}{2^3} = +$

$f(2)$

Abs Min of 8 at $x = 2$

NO Abs Max

Section 5.5 Highly Suggested Homework Problems: 1-29(odd), 43